

Bank Customer Complaint Analytics

1. Executive Summary

This project analyses customer complaint data to identify the primary drivers of customer dissatisfaction, evaluate complaint handling performance, and assess geographic complaint concentration. Using SQL (SQLite), Python, and Power BI, a hypothesis-driven analytical approach was applied to uncover complaint trends, operational bottlenecks, product-specific issues, and regional complaint patterns. The findings support data-driven decision-making aimed at improving customer experience and operational effectiveness.

2. Business Problem Statement

The bank receives a large volume of customer complaints but lacks a systematic method to analyze and monitor them. This limits the ability to identify key complaint drivers, track service performance, and implement targeted improvements, potentially leading to reduced customer satisfaction and operational inefficiencies.

3. Business Objectives

The project was designed to achieve the following objectives:

- Analyse complaint volume and trends over time.
- Identify products generating the highest number of complaints.
- Determine the most common issues and sub-issues reported by customers.
- Evaluate complaint response performance and service effectiveness.
- Identify product-specific operational bottlenecks.
- Assess geographic concentration of customer complaints.
- Develop interactive dashboards to support business decision-making.

4. Dataset Description

The analysis utilizes a titled ("**Bank of America Customer Complaints**") obtained from (**Maven Analytics**) website. While the dataset name references Bank of America, the analysis focuses on customer complaint patterns, service performance, and operational insights derived from the available records. The dataset contains customer complaint records across multiple banking products and services. Key attributes include complaint submission dates, product categories, issue and sub-issue classifications, customer locations, and response status information.

Key Fields

Category

Time
Product
Complaint Drivers
Geography
Service Performance
Volume Metrics

Fields

Date Submitted, Date Received
Product
Issue, Sub-Issue
State
Timely Response
Complaint ID

Total Records : 62,509

Products Analysed : 9

States Covered : 51

Issues & Sub-Issues : Multiple categories.

5. Methodology

A structured analytics workflow was followed to transform raw complaint records into actionable business insights.

Data Preparation

- Imported complaint records into SQLite and Python.
- Validated data quality and standardized categorical fields.
- Reviewed missing values and verified data consistency.

Exploratory Data Analysis (EDA)

- Analysed complaint trends and complaint volume distribution.
- Examined product, issue, sub-issue, and geographic patterns.
- Evaluated complaint response performance.

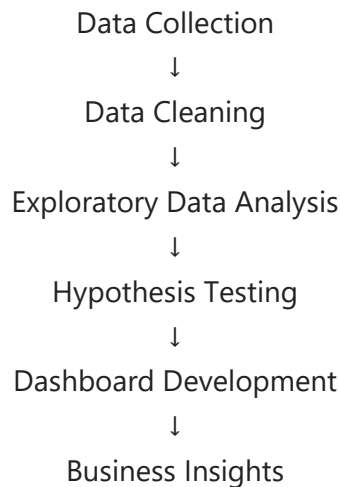
Hypothesis-Driven Analysis

- Formulated business-oriented hypotheses based on complaint behaviour.
- Performed SQL-based aggregations and comparative analysis to validate findings.

Dashboard Development

- Designed interactive Power BI dashboards to support executive monitoring, root cause analysis, operational performance evaluation, and geographic risk assessment.

Analytical Workflow



6. Analytical Hypotheses

Hypothesis 1 – Product Complaint Distribution

H0: Complaints are evenly distributed across all banking products.

H1: A small number of products account for a disproportionate share of total complaints.

Hypothesis 2 – Issue Concentration

H0: Complaint issues are uniformly distributed across categories.

H1: A limited number of issues and sub-issues drive the majority of customer complaints.

Hypothesis 3 – Response Performance

H0: Response outcomes are predominantly timely across customer complaints.

H1: Delayed and missing responses represent a meaningful portion of complaint handling outcomes.

Hypothesis 4 – Product-Level Operational Bottlenecks

H0: Response performance is consistent across all products.

H1: Certain products exhibit higher rates of delayed responses, indicating operational bottlenecks.

Hypothesis 5 – Geographic Concentration

H0: Complaint volume is evenly distributed across states.

H1: Specific states contribute a disproportionate share of total complaints.

7. Tools & Technologies

Tool	Purpose
SQL (SQLite)	Data extraction, aggregation, and hypothesis testing
Python (Pandas, NumPy, Matplotlib, Seaborn)	Data cleaning, preprocessing, and exploratory analysis
Power BI	Interactive dashboard development and business visualization
Microsoft Excel	Initial data review and validation

8. Data Analysis & Hypothesis Testing

Hypothesis 1 – Product Complaint Distribution

Finding: Complaint volume was highly concentrated within a small number of products. The top three product categories accounted for approximately 77.9% of all complaints, indicating that customer dissatisfaction was not evenly distributed across the product portfolio.

Evidence

- Checking/Savings Accounts: 24,807 complaints (39.69%)
- Credit Cards: 16,197 complaints (25.91%)
- Credit Reporting/Credit Repair Services: 7,710 complaints (12.33%)

Decision: Reject H_0 and support H_1 .

Hypothesis 2 – Issue Concentration

Finding: Complaint activity was concentrated within a limited number of issue and sub-issue categories. A small number of recurring complaint themes accounted for the majority of complaints within the highest-volume products.

Evidence

Checking/Savings Accounts

- Managing an account: 15,109 complaints (60.91%)
- Closing an account: 2,953 complaints (11.90%)
- Opening an account: 2,725 complaints (10.98%)

Credit Cards

- Problem with a purchase shown on statement: 4,415 complaints (27.26%)
- Getting a credit card: 1,867 complaints (11.53%)
- Other features, terms, or problems: 1,633 complaints (10.08%)

Credit Reporting/Credit Repair Services

- Incorrect information on credit report: 4,145 complaints (53.76%)
- Problem with investigation: 1,641 complaints (21.28%)
- Improper use of credit report: 1,517 complaints (19.68%)

Decision: Reject H_0 and support H_1 .

Hypothesis 3 – Response Performance

Finding: Complaint handling performance was generally strong, with timely responses accounting for 93.77% of all complaints. Delayed and missing responses represented a relatively small proportion of complaint outcomes.

Evidence

- Timely Responses: 58,612 complaints (93.77%)
- Delayed Responses: 2,403 complaints (3.84%)
- Missing Responses: 1,494 complaints (2.39%)

For analytical purposes, a 95% timely response rate was used as a reference benchmark. The observed performance was slightly below this benchmark, indicating opportunities for process improvement.

Decision: Support H_0 .

Hypothesis 4 – Product-Level Operational Performance

H0: Response performance is consistent across all banking products.

H1: Certain products exhibit significantly lower response performance, indicating operational bottlenecks.

Finding: Product-level SLA analysis indicates that timely response performance remained above 90% across all major banking products. While minor variations were observed between products, overall complaint-handling performance was relatively stable across the product portfolio.

Evidence

- Credit Reporting/Credit Repair Services: 91.37%
- Debt Collection: 91.89%
- Vehicle Loan or Lease: 92.26%
- Credit Card: 93.20%
- Checking/Savings Account: 93.64%
- Money Transfer Services: 93.95%
- Personal Loan: 96.40%
- Mortgage: 99.06%

Observation

As the dataset does not provide an official SLA target, a 95% timely response rate was assumed as a reference benchmark for evaluation. Under this assumption, most major products performed slightly below the benchmark. High-volume products such as Checking/Savings Accounts and Credit Cards recorded the largest numbers of delayed and missing responses, suggesting areas that may benefit from additional operational monitoring.

Decision: Support H_0 . Although performance varied slightly across products, the results do not indicate substantial operational bottlenecks or major inconsistencies in complaint response handling.

Hypothesis 5 – Geographic Concentration

Finding: Complaint volume was geographically concentrated rather than evenly distributed. A small number of states accounted for a substantial share of total complaints.

Evidence

Top 5 States:

- California: 13,708 complaints (21.93%)
- Florida: 6,486 complaints (10.38%)
- Texas: 4,686 complaints (7.50%)
- New York: 4,441 complaints (7.10%)
- Georgia: 2,921 complaints (4.67%)

The top five states contributed more than 40% of total complaints, while the top ten states accounted for over 60%.

Decision: Reject H_0 and support H_1 .

9. Key Findings

- Complaint volume was highly concentrated within a small number of products, with Checking/Savings Accounts, Credit Cards, and Credit Reporting/Credit Repair Services accounting for approximately 77.9% of all complaints.
- Customer dissatisfaction was driven by a limited number of recurring issues and sub-issues, indicating identifiable root causes rather than random complaint distribution.

- Overall complaint response performance was strong, with 93.77% of complaints receiving timely responses.
- Product-level response performance remained relatively consistent across the banking portfolio, with all major products maintaining timely response rates above 90%.
- Using an assumed 95% SLA benchmark, most products performed slightly below the target, indicating opportunities for further improvement in response efficiency.
- Complaint activity was geographically concentrated, with the top five states contributing more than 40% of total complaints and the top ten states accounting for over 60%.
- California represented the largest complaint hotspot, generating approximately 21.9% of all complaints.

10. Dashboard Design & Justification

To support business decision-making, four interactive dashboards were developed, each addressing a specific analytical objective.

Dashboard 1 – Executive Overview

Provides a high-level summary of complaint trends, product distribution, and response performance. Designed for management-level monitoring of complaint activity and service effectiveness.

Dashboard 2 – Root Cause Analysis

Focuses on complaint drivers by analysing products, issues, and sub-issues. Enables identification of recurring customer pain points and operational problem areas.

Dashboard 3 – Service Performance & Response Effectiveness

Evaluates complaint response outcomes and operational performance across products. Supports assessment of complaint handling efficiency and service quality.

Dashboard 4 – Geographic Risk & Regional Concentration

Examines complaint distribution across states and identifies geographic hotspots. Supports regional risk assessment and resource allocation decisions.

The dashboards were designed using interactive filters, drill-down capabilities, and bookmark navigation to enhance user exploration and analytical flexibility.



Bank Of America Complaint Analysis

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2.39

Missing Response in (%)

3.84

No Response in (%)

93.77

Timely Response in (%)

62.51K

Total Complaints

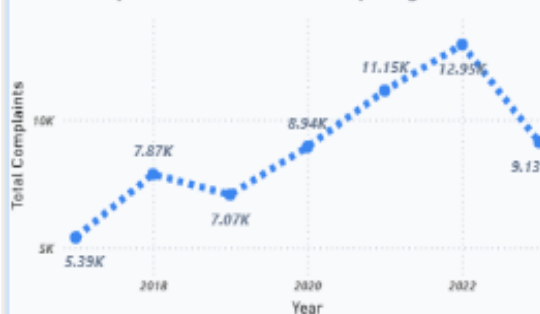
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Total Products

51

Total States

Complaint Volume Trend Across Reporting Periods



Complaint Breakdown By Submission Channel



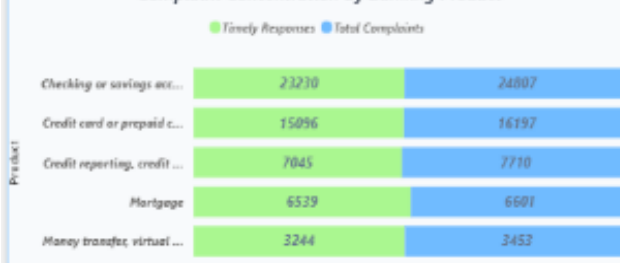
Product: All
State: All

Submitted via:
☐ Email
☐ Fax
☐ Phone
☐ Postal mail
☐ Referral

Quarter:
☐ Qtr 1
☐ Qtr 2
☐ Qtr 3
☐ Qtr 4

Year:
☐ 2017
☐ 2018
☐ 2019
☐ 2020
☐ 2021
☐ 2022

Complaint Concentration By Banking Product

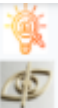


High-Impact Products Driving Complaint Volume



BANK OF AMERICA

Bank Of America Complaint Analysis



[View Detailed Root Cause](#)

Most Frequently Reported Customer Issues (Top 10 Issues)



Product

- ☐ Checking or savings account
- ☐ Credit card or prepaid card
- ☐ Credit reporting, credit repair services, or other personal consu...
- ☐ Debt collection
- ☐ Money transfer, virtual currency, or money service
- ☐ Mortgage
- ☐ Payday loan, title loan, or personal loan
- ☐ Student loan
- ☐ Vehicle loan or lease

Critical Customer Pain Points (Top 10 Sub-Issues)





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Bank Of America Complaint Analysis

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Total Complaints

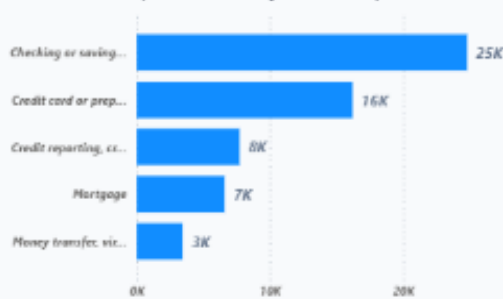
- Year
- ☐ Select all
 - ☐ 2017
 - ☐ 2018
 - ☐ 2019
 - ☐ 2020
 - ☐ 2021
 - ☐ 2022
 - ☐ 2023

- Quarter
- ☐ Select all
 - ☐ Qtr 1
 - ☐ Qtr 2
 - ☐ Qtr 3
 - ☐ Qtr 4

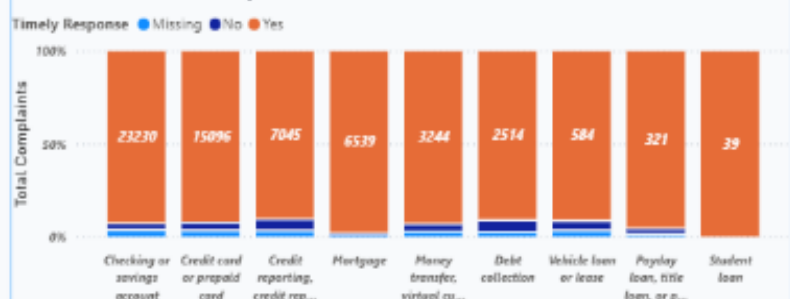
- Timely respo...
- ☐ Select all
 - ☐ Missing
 - ☐ No
 - ☐ Yes



Complaint Volume By Product (Top 5)



Response Status Distribution Across Products



Complaint Volume vs Response Effectiveness



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Bank Of America Complaint Analysis

51

Total States

21.93%

Top State Share %

51.58%

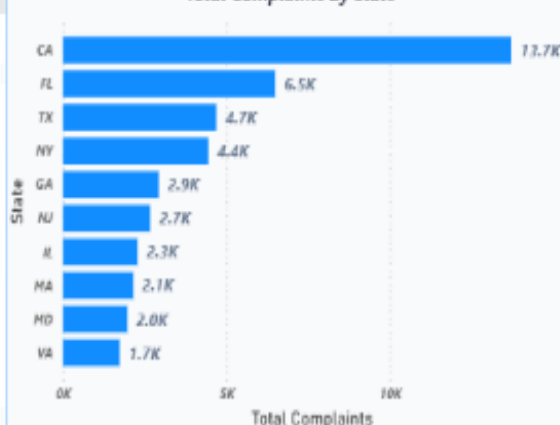
Top 5 States Share %

62.51K

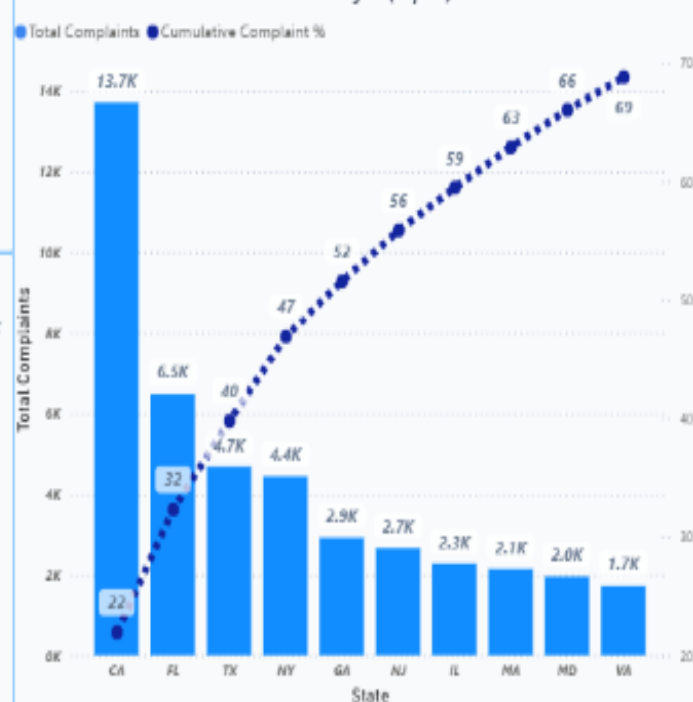
Total Complaints

- Product
- ☐ Select all
 - ☐ Checking or savings account
 - ☐ Credit card or prepaid card
 - ☐ Credit reporting, credit repair services, or other personal consumer reports
 - ☐ Debt collection
 - ☐ Money transfer, virtual currency, or money service
 - ☐ Mortgage
 - ☐ Payday loan, title loan, or personal loan
 - ☐ Student loan
 - ☐ Vehicle loan or lease

Total Complaints By State



Pareto Analysis (Top 10)



Expand Full Screen Map

Back To Default Dashboard



11. Business Recommendations

Based on the analysis, the following actions are recommended:

1. Prioritize improvement initiatives for high-complaint products, particularly Checking/Savings/Managing Accounts and Credit Cards, which account for the majority of customer complaints.
2. Address recurring complaint drivers by focusing on account management, purchase-related disputes, and credit reporting accuracy.
3. Continue monitoring complaint response performance to improve compliance with the assumed 95% service benchmark and reduce delayed response cases.
4. Allocate additional operational resources to high-volume complaint categories to minimize response backlogs and maintain service quality.
5. Investigate geographic complaint hotspots, particularly California, Florida, and Texas, to identify regional factors contributing to elevated complaint volumes.
6. Utilize dashboard monitoring as part of ongoing operational review processes to support proactive customer experience improvement.

12. Conclusion

This project successfully transformed customer complaint data into actionable business insights through SQL analysis, Python-based data preparation, and Power BI dashboard development.

The analysis revealed that complaint activity is concentrated within a small number of products, recurring issues drive a significant portion of customer dissatisfaction, and complaint volume is geographically clustered within specific states. While overall response performance remains strong, opportunities exist to further improve operational efficiency and complaint handling effectiveness.

The resulting dashboards provide decision-makers with a structured framework for monitoring complaint behaviour, identifying root causes, evaluating service performance, and prioritizing improvement initiatives. These insights can support data-driven strategies aimed at enhancing customer satisfaction and operational excellence.